

Payment & Financial Services - Document 3 - Page 1/6

Document ID: PAY-003 | Classification: Internal Use

Financial transaction security protocols: TLS 1.3, tokenization, encryption at rest and in transit.

Payment API rate limit: 100 requests/second. Use exponential backoff for 429 responses.

Payment service deployment: blue-green releases, canary deployments, and rollback procedures.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

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Payment processing performance monitoring: latency metrics, throughput, success rates, and SLA thresholds.

PCI-DSS: Never log full card numbers. Mask to last 4 digits. Secure log storage required.

Billing and invoicing: subscription management, proration, tax calculation, and invoice generation.

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Payment database management: transaction logs, audit trails, backup procedures, and data retention policies.

Webhook retry: 3 attempts at 1h, 4h, 24h. Implement idempotent handlers for duplicate events.

Refund and chargeback processing: initiation, status tracking, dispute resolution, and reason codes.

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Payment gateway configuration for EU region: PSD2 compliance, SCA requirements, and GDPR data handling.

Gateway failover: primary EU, secondary US. Health check every 30s. Auto-switch on 3 consecutive failures.

Payment processor integration: Stripe, Adyen, Braintree API documentation and migration guides.

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Payment user authentication: 3DS2 flows, biometric verification, and risk-based authentication.

Refund must be $\leq$ original amount. Partial refunds supported. Full refund voids original transaction.

Financial compliance: PCI-DSS Level 1, SOC 2, ISO 27001 requirements for payment systems.

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Payment transaction logging: structured logs, trace IDs, and observability for incident investigation.

3DS2 challenge: customer may receive OTP from bank. Timeout 5 minutes. Handle challenge completion.

Transaction reconciliation: daily settlement, mismatch resolution, and reconciliation report generation.

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